



Sultan Qaboos University

College of Economics and Political Science

Green Sukuk for Renewable Energy Infrastructure in Oman: Project Finance, Sustainability Accounting, and Islamic Capital Markets

Admission research proposal for the MSc Accounting and Finance

Applicant: Elghali Sany

Submitted: 18 May 2026

Declaration of Originality

I declare that this research proposal is my own work and has not been submitted, in whole or in part, for another degree, programme, scholarship application or academic award. All sources used in preparing this proposal have been acknowledged through appropriate citation and referencing. Any empirical analysis, financial modelling, ESG scoring, expert interview evidence or project-level validation described in this document is presented as proposed work unless explicitly stated otherwise.

Name

El Ghali Sany

Signature



Date

18 May 2026

Author Information

Author	El Ghali Sany
Academic background	Management, Finance and Accounting
Current programme	MSc 1 Procurement & Supply Chain Management
Current institution	INSEEC Lyon
Research fields	Project finance; Islamic finance; sustainability accounting; infrastructure investment
Technical orientation	DCF modelling; WACC/DSCR/LCOE analysis; ESG scorecard design; Excel/Python-based sensitivity testing
Contact	sanyghali@gmail.com
Target programme	MSc Accounting and Finance, College of Economics and Political Science, Sultan Qaboos University
Intended fit within programme	Finance and Islamic Banking and Finance pathways
Programme alignment	Research Methods; Corporate Finance and Asset Markets; Project Finance; Financial Econometrics; Islamic Capital Markets

Abstract

This research proposal treats Oman's renewable-energy build-out as a finance problem, not just a technology target. The study asks whether green sukuk can support utility-scale renewable-energy infrastructure once project cash flows, debt-service resilience, ESG evidence and Shariah structuring are tested together rather than in isolation. I position the manuscript at the proposal stage and do not present empirical results yet. Instead, I set out a replicable framework built around project-level DCF modelling, an ESG scorecard grounded in IFRS S1/S2 and the GHG Protocol, a structured comparison between green sukuk and conventional debt, and scenario-based sensitivity testing. The proposal advances three claims: first, green sukuk cannot be treated as credible infrastructure finance without project-level bankability testing; second, ESG credibility depends on auditable boundaries, baselines and reporting discipline rather than on the label alone; third, any financing advantage in Oman remains conditional and must be modelled through profit rate, tenor, fees, disclosure burden and investor-base effects. The expected output is not a promotional verdict on green sukuk. It is a decision framework that should show when the instrument strengthens project finance and when it mainly adds documentation cost without improving economics.

Keywords: green sukuk; Oman Vision 2040; renewable energy finance; ESG accounting; project finance; Islamic finance; DCF; IFRS S2; avoided emissions.

Table of Contents

Abstract	4
Methodological Note	8
1. Introduction	9
2. Research Problem, Gap, and Objectives	10
3. Literature Review	11
4. Conceptual Framework	13
5. Methodology	15
6. Expected Analytical Outputs and Academic Contribution.....	24
7. Limitations, Risks, and Mitigation	25
8. Conclusion.....	27
References	28
Appendix A. Preliminary Evidence Matrix.....	29
Appendix B. Proposed Timeline	30
Appendix C. Semi-Structured Interview Guide	32

List of Tables

Table 1. Conceptual framework linking green-sukuk structure, project-finance metrics, ESG accountability, and policy alignment for renewable-energy infrastructure in Oman.....	11
Table 2. Data requirements, source hierarchy, modelling use, and validation rules for green-sukuk renewable-energy project appraisal in Oman.	14
Table 3. Proposed ESG scorecard for assessing avoided emissions, renewable generation, social value, use-of-proceeds traceability, and governance credibility in Omani renewable-energy projects.....	17
Table 4. Analytical comparison of green sukuk and conventional debt across structure, pricing, tenor, use-of-proceeds control, reporting burden, investor base, and risk allocation.	19
Table 5. Scenario logic for testing project bankability under conservative, base-case, and transition-accelerated assumptions for green-sukuk renewable-energy finance.....	20
Table 6. Research-risk matrix assessing probability, impact, mitigation strategy, residual risk, and implementation timing for the proposed green-sukuk project-finance study.	23
Table A1. Preliminary evidence matrix mapping policy, market, disclosure, and project-finance claims to source type, reliability status, and manuscript section.	27
Table B1. Proposed twelve-month research timeline linking source validation, data collection, model construction, scenario analysis, expert validation, drafting, and final submission.....	28
Table C1. Semi-structured interview themes, target expert groups, expected evidence, and linkage to financial-model and ESG-scorecard assumptions.	30

List of Figures

Figure 1. Integrated research workflow for testing green-sukuk bankability and ESG accountability in Omani renewable-energy infrastructure.	14
Figure 2. Discounted cash-flow model architecture linking project assumptions to NPV, IRR, DSCR, LCOE, and bankability interpretation.	18
Figure 3. Decision matrix crossing financial bankability and ESG credibility for green-sukuk-financed renewable-energy projects in Oman.	20
Figure 4. Illustrative tornado chart ranking the expected sensitivity of project NPV to financing cost, CAPEX, tariff, capacity factor, OPEX, and carbon-value assumptions.....	23
Figure 5. Probability-impact matrix of research-design risks affecting data access, sukuk-pricing evidence, expert validation, and model complexity.	26
Figure B1. Twelve-month Gantt chart sequencing source validation, data collection, DCF modelling, ESG scorecard construction, expert validation, drafting, and final submission.....	31

1. Introduction

This proposal tests whether green sukuk can finance renewable-energy infrastructure in Oman once bankability, ESG evidence and Shariah structuring are examined in one project-level framework. The core question is narrow. Can the instrument survive financial scrutiny, disclosure scrutiny and governance scrutiny at the same time?

The scale is concrete. Official Omani communications refer to a 30% renewable-electricity target by 2030 and describe a pipeline of solar and wind assets that would reshape the generation mix. Demand is rising at the same time. That combination turns the energy transition into a capital-allocation problem: CAPEX, tariff design, capacity factor, debt service and disclosure cost will decide whether the assets remain investable.

Public finance cannot carry the full transition. Renewable infrastructure is capital intensive at the start, then relatively cheap to operate, which makes the financing rate unusually powerful in the final economics. A small movement in WACC can change NPV, DSCR and tariff affordability. That is why a green financing instrument must be tested inside a project-finance model, not praised in abstract policy language.

Green sukuk deserve attention because they sit exactly at that pressure point. They can connect Shariah-sensitive capital with labelled green assets, but the label does not settle the financing question. A sukuk can be Shariah-compliant and green-labelled while still failing as infrastructure finance if DSCR collapses under CAPEX, tariff or profit-rate stress.

The gap is not the absence of literature on green sukuk, renewable project finance or ESG disclosure. The gap is their separation. Green sukuk studies often stop at instrument legitimacy and market development. Renewable-finance studies test cash-flow resilience but rarely model Islamic capital-market structures. ESG research debates disclosure and avoided-emissions boundaries, yet it seldom connects reporting discipline to WACC, DSCR or investor appetite in a single Omani project case.

This proposal therefore tests three claims rather than presenting green sukuk as an automatic solution.

1. Green sukuk require project-level DSCR and WACC testing before they can be treated as bankable infrastructure finance.
2. ESG credibility depends on auditable baselines, boundaries, emission factors and reporting governance, not on the green label itself.
3. The effect of green sukuk on WACC in Oman is uncertain and must be modelled through profit rate, tenor, issuance cost, investor base and reporting burden.

2. Research Problem, Gap, and Objectives

2.1 Research problem

The problem is narrow but demanding: Oman lacks a project-level framework for testing whether green sukuk can finance utility-scale renewable infrastructure while preserving bankability and ESG credibility. A policy target can support green finance, and a sukuk can carry a green label, but neither tells investors whether cash flows cover debt service, whether avoided emissions are measured properly, or whether disclosure costs change the economics.

2.2 Research gap

The gap has three layers. Green sukuk studies give useful insight into Shariah compliance, investor appetite and market development, but they rarely build project-level cash-flow models. Renewable-energy finance studies work with DCF, WACC and LCOE, yet often treat Islamic capital-market instruments as outside the model. ESG literature gives disclosure rules, including IFRS S1/S2 and avoided-emissions logic, but usually stops before translating them into a weighted project scorecard. Oman sits precisely where those literatures fail to meet.

2.3 Main research question

How can a project-level framework test the bankability and ESG credibility of green sukuk-financed renewable-energy infrastructure in Oman?

2.4 Sub-questions

4. What financial profile does a representative solar PV or wind infrastructure project in Oman produce when tested through NPV, IRR, WACC, LCOE, payback and DSCR?
5. Which ESG indicators can credibly capture climate impact, local value, governance discipline and disclosure quality for sukuk-financed renewable-energy projects?
6. How do the structural features of green sukuk - asset-backing, profit rate, maturity, use-of-proceeds rules and reporting duties - compare with conventional debt?
7. How sensitive is project bankability to energy-price assumptions, carbon-value assumptions, CAPEX, capacity factor and financing cost?
8. Which policy and investor practices would make green sukuk more credible and scalable for Oman's renewable-energy pipeline?

2.5 Objectives

9. Build a DCF-based financial model for a representative renewable-energy project in Oman.
10. Construct an ESG scorecard aligned with IFRS S1/S2 and the GHG Protocol approach to avoided emissions.
11. Compare green sukuk and conventional debt through cost of capital, reporting burden, risk allocation and strategic fit.
12. Run sensitivity and scenario tests for financing cost, CAPEX, tariff, energy output, carbon value and exchange-rate assumptions.
13. Draft recommendations for regulators, Islamic banks, investors and project sponsors.

2.6 Boundary conditions and scope

This framework applies to utility-scale renewable-energy infrastructure in Oman, with solar PV and onshore wind as the preferred reference technologies. It does not evaluate small rooftop systems, corporate sukuk unrelated to traceable assets, or labelled instruments without use-of-proceeds and impact-reporting discipline. The study treats Shariah compliance and Vision 2040 alignment as boundary conditions for interpreting the financing result, not as substitutes for project-level bankability.

3. Literature Review

3.1 Oman's energy transition and sustainable finance context

Oman's energy transition is shaped by three pressures: diversification, energy security and climate policy. The Ministry of Foreign Affairs reports a 30% renewable-electricity target by 2030 and links new solar and wind projects to security of supply and sustainability. In practice, those targets turn into financing choices: who funds grid-connected renewable assets, how risks are allocated and how climate benefits are measured.

The finance architecture is moving as well. Reuters reported that Oman's 2024 Sustainable Finance Framework was designed to reduce fossil-fuel dependence, attract ESG investors and support instruments such as green, social and sustainability-linked bonds or sukuk. That framework matters for this proposal because it gives legal and market context to a question that would otherwise remain abstract: how can a labelled Islamic instrument finance physical energy infrastructure without weakening disclosure discipline?

3.2 Green sukuk and sustainable Islamic finance

Green sukuk sit at the intersection of Islamic finance and use-of-proceeds sustainability finance. Ulfah et al. (2023) show how the literature frames them as vehicles for renewable energy, climate-aligned projects and SDG-linked investment. That framing is useful, but it often stops too early. Instrument legitimacy does not prove project bankability; a Shariah-compliant green sukuk can still fail if the asset cannot absorb CAPEX stress, tariff uncertainty or profit-rate pressure.

Oman's first green sukuk gives the topic a market anchor. Oman Sustainability Week reported that Oman Electricity Transmission Company issued a USD 750 million five-year green sukuk in October 2025, with proceeds allocated to eligible green projects. This case does not replace project-level modelling, but it shows that Oman has crossed from policy language into capital-market execution.

3.3 ESG accounting and climate disclosure

ESG credibility determines whether green sukuk remain finance instruments or drift into greenwashing risk. IFRS S1 requires disclosure of material sustainability-related financial risks and opportunities; IFRS S2 narrows that discipline to climate-related matters. For this proposal, the standards help translate climate exposure, transition risk and governance duties into variables that a project sponsor or sukuk issuer must document.

For emissions impact, the GHG Protocol's avoided-emissions guidance offers a usable but demanding basis. It forces the researcher to state the baseline, boundary and emission factor instead of claiming a clean-energy benefit in broad language. That discipline matters in Oman, where renewable generation must be compared with the grid it displaces, not with an imaginary zero-emission benchmark.

3.4 Renewable energy project finance

Renewable-energy projects concentrate their economics in upfront CAPEX, financing cost and generation performance. IRENA's 2024 cost data confirm that renewables remain cost-competitive in many markets, but cost competitiveness at global level does not automatically make a single Omani project bankable. The project must still carry local assumptions on tariff, grid access, capacity factor, currency exposure and financing terms.

DCF modelling fits the research because it makes those assumptions visible. NPV shows whether discounted cash flows exceed the initial investment. IRR gives the project's break-even discount rate. WACC captures the cost of blended capital. LCOE translates lifecycle cost into a unit electricity cost. DSCR tests the project's capacity to serve debt or sukuk obligations year by year. Together, these metrics prevent the proposal from treating green finance as a slogan.

3.5 Synthesis of the gap

Taken together, the literature justifies the topic but leaves a methodological gap. Green sukuk, ESG disclosure and renewable project finance rarely meet inside one project-level test. Most studies stop too early: they explain the instrument, the policy context or the disclosure logic, but they do not show how financing cost, DSCR resilience, emissions accounting and reporting burden interact inside the same infrastructure case. This proposal is designed to close that gap in a way that can be falsified, replicated and challenged.

The review therefore separates the literature into four streams and uses each stream for a specific test: green sukuk literature explains instrument legitimacy, renewable-project finance explains cash-flow resilience, ESG disclosure literature explains the auditability of climate claims, and Oman/GCC policy sources define the local boundary conditions. The contribution is the integration of these streams into one project-level test, not another descriptive account of green finance.

4. Conceptual Framework

The conceptual framework treats sukuk design as a mechanism that affects financing cost, risk allocation and disclosure burden, not as a green label attached after the project is already bankable. Financing structure shapes WACC, debt-service burden, reporting duties and investor base. Those variables then affect NPV, IRR, DSCR, LCOE and ESG credibility. The logic is intentionally sequential: no financing structure should be called successful unless it survives project-level testing and evidence-based impact reporting.

Table 1. Conceptual framework linking green-sukuk structure, project-finance metrics, ESG accountability, and policy alignment for renewable-energy infrastructure in Oman.

Construct	Core variables	Measurement approach	Expected output
Financing structure	Green sukuk type, profit rate, maturity, fees, use-of-proceeds rules	Document analysis and market data comparison	Cost-of-capital and compliance profile
Project economics	CAPEX, OPEX, tariff/revenue, capacity factor, degradation, inflation	DCF model over 20–25 years	NPV, IRR, payback, DSCR
Energy performance	Annual generation, capacity, curtailment, storage integration	Technical assumptions from PWP, IRENA, project documents	MWh/year and LCOE
Environmental impact	Grid emission factor, renewable output, avoided emissions	GHG Protocol comparative emissions logic	tCO ₂ e avoided per year and lifecycle
Social and governance impact	Local employment, procurement transparency, Shariah governance, external review	ESG scorecard and qualitative review	Composite ESG credibility score
Policy alignment	Vision 2040, net-zero 2050, sustainable finance framework	Policy-document mapping	Strategic alignment assessment

Note. The framework is used as the organising map for the proposal: financing structure shapes cost of capital and risk allocation, while project economics and ESG evidence determine whether the instrument is credible. Abbreviations: DCF, discounted cash flow; DSCR, debt service coverage ratio; ESG, environmental, social, and governance; GHG, greenhouse gas; IRR, internal rate of return; LCOE, levelised cost of electricity; NPV, net present value; WACC, weighted average cost of capital.

Figure 1 turns the conceptual framework into a workflow, so the reader can follow the path from policy context to data validation, financial modelling, ESG scoring, sukuk comparison and recommendations.

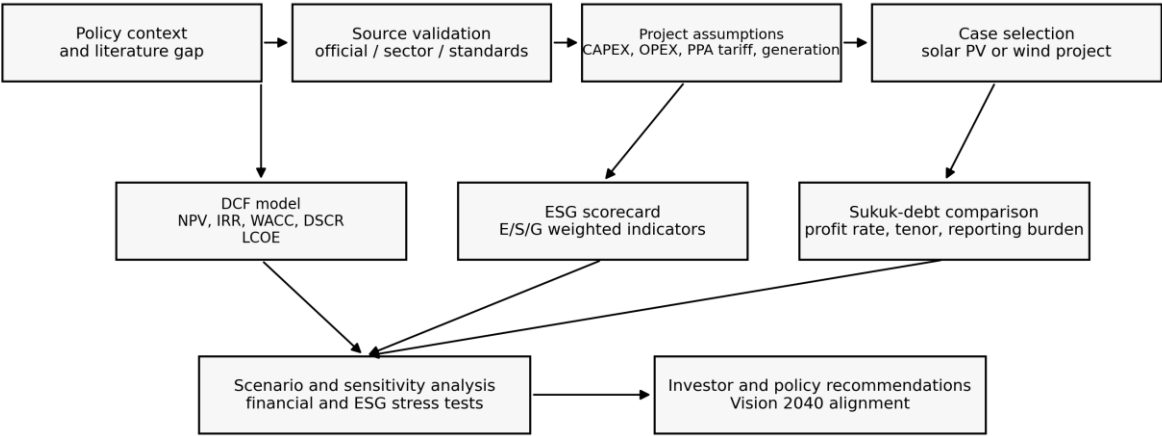


Figure 1. Integrated research workflow for testing green-sukuk bankability and ESG accountability in Omani renewable-energy infrastructure.

Note. The workflow follows the proposed sequence from policy context and data collection to financial modelling, ESG scoring, sukuk comparison, sensitivity testing, and recommendations. Abbreviations: DCF, discounted cash flow; ESG, environmental, social, and governance.

5. Methodology

5.1 Research design

I use a mixed-methods design because the research problem cannot be resolved by narrative interpretation alone. The quantitative module tests bankability through DCF modelling. The ESG module checks whether climate and governance claims can be supported by auditable evidence. The qualitative module does not replace the model; it tests assumptions, identifies hidden execution risks and challenges weak inputs before they harden into conclusions.

5.2 Unit of analysis and case selection

The unit of analysis is a utility-scale renewable-energy project in Oman, preferably solar PV or onshore wind because both technologies appear in the national project pipeline. The OETC green sukuk serves as a market case for structuring analysis; it does not replace project-level modelling. If confidential project data remain inaccessible, I will triangulate public information from Nama PWP, Omani energy authorities, IRENA benchmarks and available green-finance documentation.

5.3 Model calibration protocol

The model will use a 25-year base-case horizon unless project documentation justifies another operating life. The base currency will be USD, with OMR sensitivity where exchange-rate or local-cost assumptions materially affect the result. Discounting will use a nominal post-tax WACC unless data availability requires a pre-tax proxy; in that case, the limitation will be stated in the model notes. Revenue will be modelled through an explicit tariff or PPA assumption, generation through installed capacity, capacity factor, annual hours and degradation adjustment, and debt or sukuk service through tenor, profit/coupon rate, fees and repayment profile. Bankability will be judged through a stated DSCR covenant threshold before scenario testing begins. Avoided emissions will be calculated from renewable generation, grid-emission factor and project-emission boundary; no avoided-emissions result will be accepted without a disclosed baseline and boundary.

Each formula therefore has four controls: input source, assumption range, decision use and sensitivity test. This prevents the equations from becoming math-washing. The contribution is not the use of standard NPV or WACC formulas; it is their calibration to Oman's renewable-energy and green-sukuk context.

5.4 Data sources

Table 2. Data requirements, source hierarchy, modelling use, and validation rules for green-sukuk renewable-energy project appraisal in Oman.

Data item	Preferred source	Access status / frequency	Model use	Limitation / validation rule
Project CAPEX and OPEX	IRENA cost reports; Oman project documents; Nama PWP material where available	Public / annual or project-specific	Build investment, operating-cost, and lifecycle cash-flow assumptions	Use Oman-specific data first; otherwise triangulate regional/global cost benchmarks
Tariff, PPA, and electricity-demand assumptions	Nama PWP statements; sector regulator publications; official Omani communications	Public / annual planning documents	Estimate revenues, DSCR, and bankability under base and stress scenarios	Clearly separate observed tariff data from scenario assumptions
Financing cost and sukuk terms	OETC green finance/sukuk documentation; market reports; official framework documents	Public / issuance-specific	Compare sukuk profit rate, tenor, and reporting cost against conventional debt	Avoid claiming cost-of-capital reduction unless supported by market evidence
Emission factors and avoided emissions	GHG Protocol; national grid or electricity-sector estimates where available	Public / methodology-dependent	Calculate tCO ₂ e avoided by renewable generation relative to baseline	Disclose boundary, baseline, and emission factor to avoid greenwashing risk
ESG disclosure standards	IFRS S1/S2, GHG Protocol, and relevant sector reports	Public / standards-based	Define ESG indicators and reporting requirements	Distinguish mandatory disclosure requirements from voluntary best practice
Expert validation evidence	Semi-structured interviews with finance, energy, ESG, and policy experts	Primary / subject to access and ethics approval	Validate weighting, risk assumptions, and feasibility constraints	Use interview data as triangulation, not as a substitute for verified quantitative inputs

Note. The protocol prioritizes Oman-specific evidence; regional or international benchmarks serve as proxies only when project-level data are unavailable. Abbreviations: CAPEX, capital expenditure; DSCR, debt service coverage ratio; IRENA, International Renewable Energy Agency; OPEX, operating expenditure; PPA, power purchase agreement; PWP, Power and Water Procurement; tCO₂e, tonnes of carbon dioxide equivalent.

5.5 Financial modelling equations

The model should be built in Excel or Python with a transparent assumptions sheet, scenario controls and an audit trail for every input. These equations form the financial core, but each one must be tied to a source hierarchy, sensitivity range and decision rule:

Net Present Value (NPV): $NPV = \sum_{t=1}^n FCF_t / (1 + WACC)^t - \text{Initial CAPEX}$

A positive NPV means the project clears the required return after discounting cash flows.

Internal Rate of Return (IRR): $0 = \sum_{t=1}^n FCF_t / (1 + IRR)^t - \text{Initial CAPEX}$

IRR is compared with WACC and investor hurdle rates.

Weighted Average Cost of Capital (WACC): $WACC = (E/V \times R_e) + (D/V \times R_d \times (1 - T))$

For sukuk, R_d is replaced by the sukuk profit rate, then adjusted for issuance cost and liquidity where the data justify it.

Levelised Cost of Electricity (LCOE): $LCOE = \sum_{t=0}^n \text{Costs}_t / (1 + r)^t \div \sum_{t=1}^n \text{Electricity}_t / (1 + r)^t$

LCOE measures discounted lifecycle cost per kWh or MWh generated.

Debt Service Coverage Ratio (DSCR): $DSCR_t = \text{Cash Flow Available for Debt Service}_t \div \text{Debt Service}_t$

A DSCR below lender or sukuk-holder thresholds signals refinancing pressure or default risk.

Avoided emissions: $\text{Avoided CO2e}_t = \text{Renewable electricity}_t \times \text{Grid emission factor}_t - \text{Project emissions}_t$

The boundary, baseline and emission factor must be disclosed; otherwise, avoided-emissions claims can be overstated.

Figure 2 shows how the financial model converts assumptions into cash-flow outputs and bankability metrics. The formulas are not decorative; they form the route from input data to decision criteria.

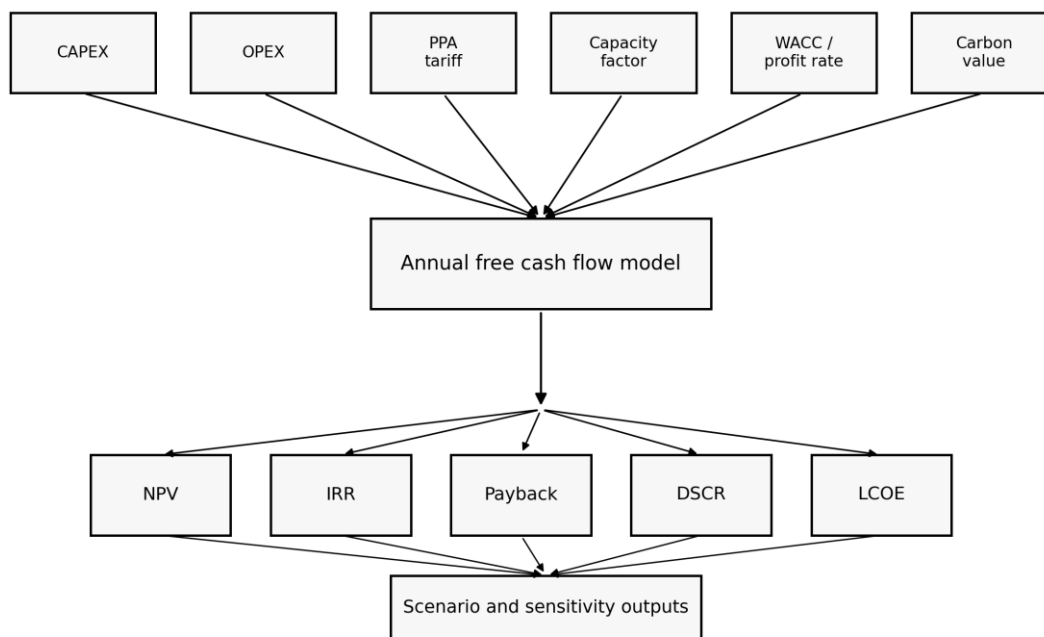


Figure 2. Discounted cash-flow model architecture linking project assumptions to NPV, IRR, DSCR, LCOE, and bankability interpretation.

Note. The architecture converts project assumptions into cash-flow outputs and bankability metrics. Abbreviations: CAPEX, capital expenditure; DSCR, debt service coverage ratio; IRR, internal rate of return; LCOE, levelised cost of electricity; NPV, net present value; OPEX, operating expenditure; PPA, power purchase agreement; WACC, weighted average cost of capital.

5.6 ESG scorecard

The ESG scorecard should remain weighted but usable. I propose a 50% environmental, 25% social and 25% governance structure, then test those weights through expert review or sensitivity analysis. Each criterion is scored from 0 to 5, where 0 means no usable evidence and 5 means documented, project-specific and externally verifiable evidence.

Table 3. Proposed ESG scorecard for assessing avoided emissions, renewable generation, social value, use-of-proceeds traceability, and governance credibility in Omani renewable-energy projects.

Dimension	Indicator	Measurement	Weight	Scoring rule	Evidence required
Environmental	Avoided emissions	tCO ₂ e/year versus baseline	20%	0 = no estimate; 5 = third-party-verifiable baseline and emission factor	Generation data, baseline, emission factor, and calculation boundary
Environmental	Renewable generation	MWh/year delivered to grid	10%	0 = unavailable; 5 = verified generation forecast or operational data	Plant capacity, capacity factor, grid-delivery estimate
Environmental	Resource efficiency	Water/material footprint and land-use considerations	10%	0 = no disclosure; 5 = quantified and mitigated project impacts	Technical studies or project environmental documentation
Environmental	Climate-risk resilience	Exposure to heat, grid, curtailment, and extreme-weather risks	10%	0 = not assessed; 5 = quantified resilience measures	Risk assessment and adaptation assumptions
Social	Local employment and skills	Jobs, training, and local value creation	15%	0 = no local-benefit evidence; 5 = measured local employment and training plan	Project staffing plan or expert validation
Social	Community and affordability relevance	Stakeholder impacts and energy-access contribution	10%	0 = no stakeholder analysis; 5 = documented stakeholder and affordability assessment	Stakeholder documentation or interviews
Governance	Use-of-proceeds and allocation reporting	Traceability of sukuk proceeds to eligible assets	15%	0 = no allocation logic; 5 = audited allocation and impact reporting	Green finance framework and reporting template
Governance	Shariah and ESG governance	Shariah board review, ESG governance, and external assurance	10%	0 = no governance structure; 5 = clear governance and external review	Sukuk documentation and assurance evidence

Note. Weights are provisional and should be tested through expert review or sensitivity analysis before final scoring. The 0-5 scale measures evidence quality, from no usable evidence to project-specific and externally verifiable evidence. Abbreviations: ESG, environmental, social, and governance; MWh, megawatt-hour; tCO₂e, tonnes of carbon dioxide equivalent.

Figure 3 translates the ESG scorecard into an investment-decision matrix. It separates projects that are bankable, ESG-credible, both, or neither.

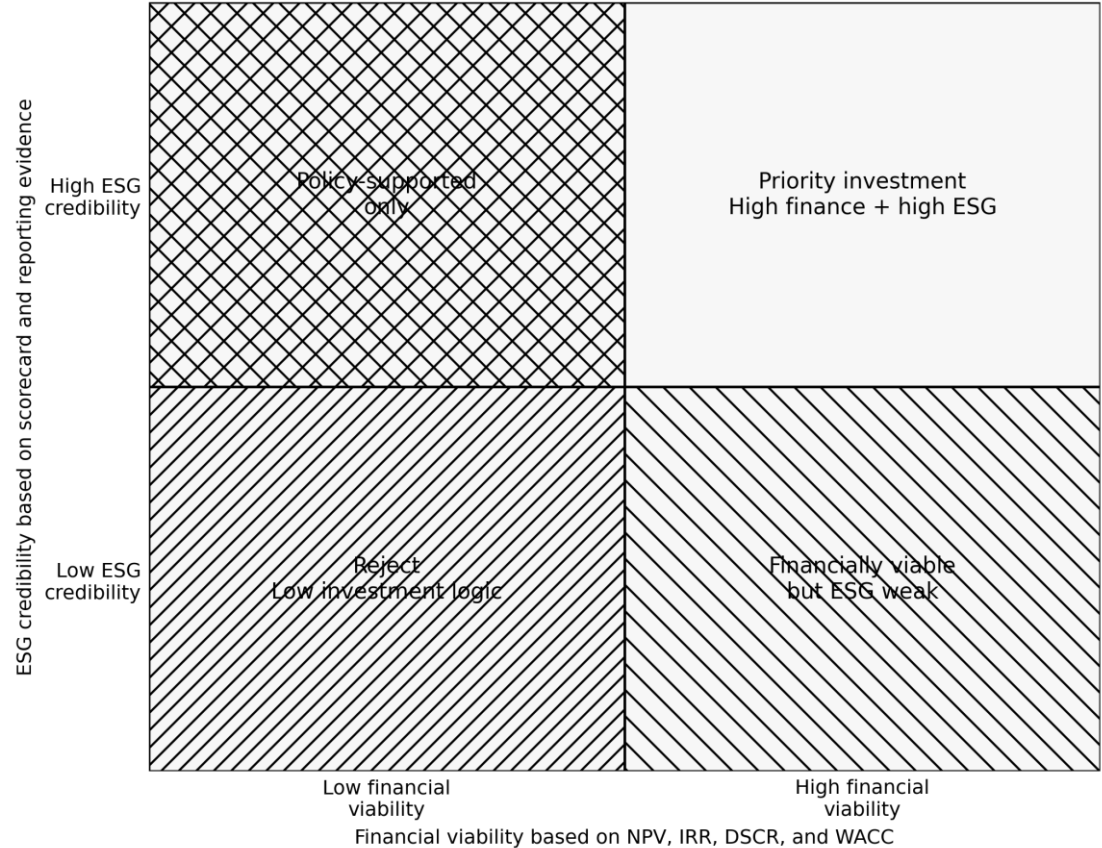


Figure 3. Decision matrix crossing financial bankability and ESG credibility for green-sukuk-financed renewable-energy projects in Oman.

Note. The matrix separates projects by financial bankability and ESG credibility to show whether a project is investable, credible, both, or neither. Abbreviations: DSCR, debt service coverage ratio; ESG, environmental, social, and governance; IRR, internal rate of return; LCOE, levelised cost of electricity; NPV, net present value; WACC, weighted average cost of capital.

5.7 Sukuk versus conventional debt comparison

The sukuk comparison cannot stop at 'Islamic versus non-Islamic'. The analysis must focus on economic and governance differences: asset-backing, investor base, profit-rate structure, tenor, issuance costs, reporting duties, Shariah compliance and refinancing risk. The model should test whether green sukuk changes WACC through investor diversification, ESG demand or pricing advantage. If no pricing benefit appears, the study should say so plainly instead of assuming a green premium.

Table 4. Analytical comparison of green sukuk and conventional debt across structure, pricing, tenor, use-of-proceeds control, reporting burden, investor base, and risk allocation.

Criterion	Green sukuk	Conventional debt/bond	Quantitative proxy	Research test
Legal/economic basis	Asset-backed or asset-based Islamic structure, often Ijara or Wakalah	Debt claim with contractual interest/coupon	Documentation cost; asset-transfer or usufruct cost	Does structure affect transaction complexity and issuance cost?
Return mechanism	Profit rate linked to sukuk structure	Coupon or loan interest rate	Profit rate/coupon spread; all-in financing cost	Does sukuk pricing change WACC relative to conventional debt?
Tenor and repayment profile	Tenor depends on investor appetite and eligible asset pool	Tenor depends on lender/bond market conditions	Maturity; grace period; amortisation profile	Which structure best supports long-lived renewable assets?
Use-of-proceeds control	Restricted to eligible green assets under framework	May be general corporate or project-specific	Share of proceeds allocated to eligible assets	Does green labelling improve traceability?
Reporting obligation	Allocation and impact reporting expected	Lower ESG-reporting burden unless green-labelled	Annual reporting cost; assurance cost	Do reporting costs reduce or enhance investment credibility?
Investor base	Islamic and ESG-oriented investors	Conventional fixed-income and bank lenders	Investor demand; subscription ratio; secondary liquidity	Does sukuk diversify the investor base?
Risk allocation	Asset and project risk shaped by sukuk structure	Credit risk governed by loan/bond covenants	DSCR covenant; reserve accounts; default triggers	Which structure creates stronger bankability discipline?

Note. The comparison treats sukuk and conventional debt as financing structures with different pricing, tenor, reporting, investor-base, and risk-allocation implications. Abbreviations: DSCR, debt service coverage ratio; ESG, environmental, social, and governance; WACC, weighted average cost of capital.

5.8 Scenario and sensitivity analysis

The study should run three scenarios at minimum: conservative, base and transition-accelerated. The sensitivity tests should cover CAPEX, capacity factor, WACC or sukuk profit rate, tariff, OPEX, carbon value and exchange-rate assumptions. The minimum outputs are a tornado chart for NPV drivers, a DSCR-by-year table, a WACC sensitivity matrix and an ESG-financial scorecard.

Table 5. Scenario logic for testing project bankability under conservative, base-case, and transition-accelerated assumptions for green-sukuk renewable-energy finance.

Variable shocked	Conservative case	Base case	Transition-accelerated case	Expected output
WACC / sukuk profit rate	Higher financing cost and lower investor appetite	Market-consistent financing cost	Lower spread through stronger green/Islamic investor demand	NPV, IRR, DSCR sensitivity
CAPEX	Cost overrun or higher EPC prices	Benchmark utility-scale CAPEX	Cost decline from procurement and technology learning	Payback and LCOE sensitivity
Capacity factor	Lower generation due to technical/grid constraints	Expected under reference assumptions	Improved dispatch and operational availability	Revenue and avoided-emissions sensitivity
Tariff / PPA revenue	Lower tariff or delayed offtake certainty	Reference PPA/tariff assumption	Improved tariff certainty and contractual bankability	DSCR and payback sensitivity
Carbon value / climate benefit	No carbon value monetised	Disclosure-only avoided-emissions estimate	Policy or market value attached to verified avoided emissions	ESG-finance scenario interpretation

Note. The scenarios test how bankability changes under conservative, base-case, and transition-accelerated assumptions. Abbreviations: CAPEX, capital expenditure; DSCR, debt service coverage ratio; IRR, internal rate of return; LCOE, levelised cost of electricity; NPV, net present value; OPEX, operating expenditure; PPA, power purchase agreement; WACC, weighted average cost of capital.

Figure 4 shows the expected sensitivity-output format. The values are placeholders and must be replaced once the project data and financing assumptions are calibrated.

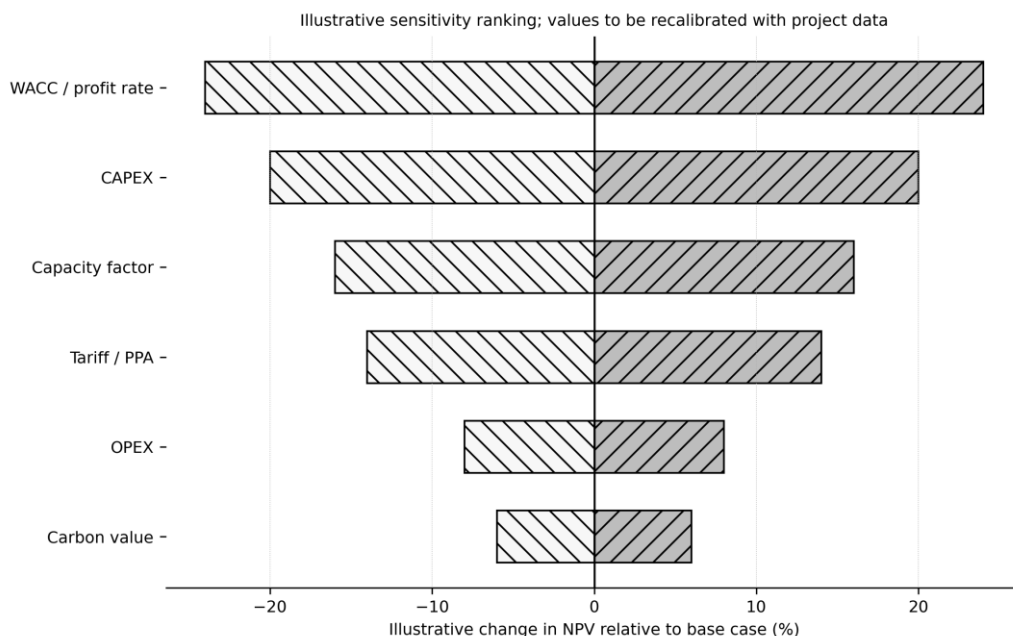


Figure 4. Illustrative tornado chart ranking the expected sensitivity of project NPV to financing cost, CAPEX, tariff, capacity factor, OPEX, and carbon-value assumptions.

Note. The tornado chart format ranks the assumptions expected to have the largest effect on project NPV after model calibration. Abbreviations: CAPEX, capital expenditure; NPV, net present value; OPEX, operating expenditure; PPA, power purchase agreement; WACC, weighted average cost of capital.

5.9 Expert validation and qualitative analysis

The qualitative component will use purposive expert sampling rather than convenience commentary. The target sample is 12-16 semi-structured interviews: 3-4 Islamic-finance or banking experts, 3-4 energy-sector or project-finance experts, 3-4 ESG/reporting specialists, and 3-4 policy or regulatory informants. Interviews should last 30-45 minutes and may be conducted online or in person. The protocol will use the themes in Table C1 as predefined coding categories, while allowing open coding for risks that the document review and DCF model do not capture.

Interview evidence will not be treated as proof of financial viability. It will be used to validate assumptions, identify missing risks, interpret sukuk documentation and test whether the proposed ESG scorecard is plausible for Omani renewable-energy projects. Findings from interviews will be triangulated against official documents, market disclosures and model outputs.

6. Expected Analytical Outputs and Academic Contribution

6.1 Expected analytical outputs

- A calibrated DCF workbook that reports NPV, IRR, LCOE, payback and DSCR under conservative, base and transition-accelerated assumptions.
- A sukuk-versus-debt comparison that shows whether green sukuk lowers, raises or leaves WACC broadly unchanged once profit rate, tenor, fees, investor base and reporting burden are modelled together.
- A sensitivity output that identifies the variables that dominate bankability under Omani conditions, especially financing cost, CAPEX, capacity factor, tariff design and carbon-value assumptions.
- An ESG scorecard that translates IFRS S1/S2 logic and GHG Protocol avoided-emissions discipline into project-level evidence rather than generic sustainability language.
- A decision rule for policymakers and investors showing when green sukuk improve credibility and when they mainly add documentation cost without improving project economics.

The expected output is not a promotional yes-or-no answer on green sukuk. The stronger result is conditional. The framework should reveal the combinations of tariff, CAPEX, financing cost, generation performance and disclosure burden under which green sukuk improve project quality—and the combinations under which they remain symbolically attractive but economically weak.

6.2 Academic contribution

The academic contribution lies in shifting the debate from instrument description to project-level testing. I do not treat green sukuk as a legal wrapper automatically aligned with sustainable infrastructure. I treat it as a financing structure that must clear the same bankability discipline as any other long-lived capital instrument, while also surviving ESG and disclosure scrutiny.

6.3 Practical contribution

The practical contribution is direct. Policymakers can use the framework to refine taxonomy design, disclosure expectations and market-development priorities. Investors and arrangers can use it to distinguish between projects that are genuinely financeable and those that rely on the green label to mask unresolved cash-flow or reporting weaknesses. Project sponsors can use it to identify which assumptions deserve early negotiation rather than late-stage repair.

6.4 Falsification logic

The framework may show that green sukuk do not reduce WACC under some scenarios. That result would not weaken the study. It would clarify that the instrument's value may lie in investor-base diversification, Shariah alignment or disclosure discipline rather than in cheaper capital. A credible framework must allow that outcome instead of assuming success in advance.

7. Limitations, Risks, and Mitigation

Table 6. Research-risk matrix assessing probability, impact, mitigation strategy, residual risk, and implementation timing for the proposed green-sukuk project-finance study.

Risk or limitation	Probability	Impact	Mitigation strategy	Residual risk	Owner / timing
Missing Oman-specific CAPEX/OPEX data	High	High	Triangulate IRENA, Nama project reports, and regional benchmarks	Medium	Researcher, months 1-3
Limited access to market pricing for sukuk and debt	Medium	High	Use public issuance data, Reuters/market reports, and expert validation	Medium	Researcher, months 2-6
Regulatory or policy changes during the study	Medium	Medium	Frame scenarios around current policy and document any changes transparently	Low-Medium	Researcher, continuous
Limited expert-interview access	High	Medium	Use multiple contact channels and a document-analysis fallback	Medium	Researcher/supervisor, months 6-8
Model complexity and over-parameterisation	Medium	Medium	Build modular DCF, ESG, and sukuk-comparison components before integration	Low-Medium	Researcher, months 3-7
Greenwashing or overstated avoided-emissions claims	Medium	High	Disclose baseline, boundaries, and emission factor; avoid unverified impact claims	Low-Medium	Researcher, months 4-7

Note. Risk levels are planning judgments and should be updated after source validation, data-access confirmation, market-document review, and expert availability checks. Abbreviations: CAPEX, capital expenditure; IRENA, International Renewable Energy Agency; OPEX, operating expenditure; PWP, Power and Water Procurement.

Figure 5 condenses the risk table into a severity matrix, making feasibility risks visible by probability and impact rather than burying them in a narrative list.

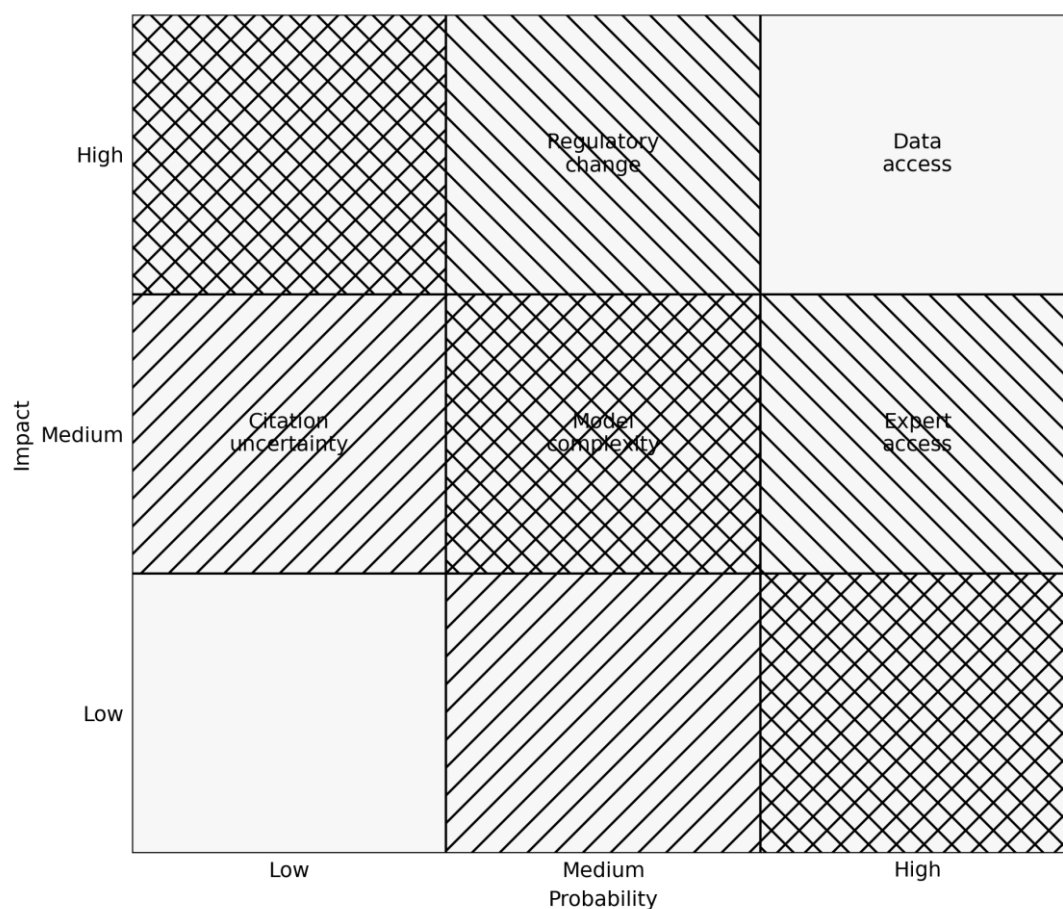


Figure 5. Probability-impact matrix of research-design risks affecting data access, sukuk-pricing evidence, expert validation, and model complexity.

Note. The matrix positions the main feasibility risks by probability and impact, with emphasis on data access, sukuk-pricing evidence, expert validation, model complexity, and greenwashing risk. Abbreviations: CAPEX, capital expenditure; OPEX, operating expenditure.

8. Conclusion

Oman's renewable-energy transition creates a precise research opportunity at the intersection of project finance, sustainability accounting and Islamic capital markets. This proposal does not assume that green sukuk are inherently superior to conventional debt. It sets up a disciplined test. If the model shows that the instrument improves bankability and ESG credibility under identifiable conditions, that conclusion will be useful. If it shows the opposite, that result will be just as valuable.

References

- Al-Shidhani, M. A., & Hayyan, M. (2025). Oman's energy transition roadmap to net zero 2050. *Energy Nexus*. <https://doi.org/10.1016/j.nexus.2025.100493>
- Baker Botts. (2025). Baker Botts advises on first green sukuk by an Omani entity. <https://www.bakerbotts.com/news/2025/10/baker-botts-advises-on-first-green-sukuk-by-an-omani-entity>
- Climate Change Laws of the World. (2024). Oman's Sustainable Finance Framework. Grantham Research Institute on Climate Change and the Environment. https://climate-laws.org/document/oman-s-sustainable-finance-framework_18ea
- GHG Protocol. (n.d.). Estimating and reporting avoided emissions. <https://ghgprotocol.org/estimating-and-reporting-avoided-emissions>
- IFRS Foundation. (2023). IFRS S1 General Requirements for Disclosure of Sustainability-related Financial Information. <https://www.ifrs.org/issued-standards/ifrs-sustainability-standards-navigator/ifrs-s1-general-requirements/>
- IFRS Foundation. (2023). IFRS S2 Climate-related Disclosures. <https://www.ifrs.org/issued-standards/ifrs-sustainability-standards-navigator/ifrs-s2-climate-related-disclosures/>
- International Renewable Energy Agency. (2025). Renewable power generation costs in 2024. <https://www.irena.org/Publications/2025/Jun/Renewable-Power-Generation-Costs-in-2024>
- Ministry of Finance, Sultanate of Oman. (2024). Sustainable Finance Framework. <https://www.mof.gov.om/>
- Ministry of Foreign Affairs, Sultanate of Oman. (2024). Oman launches Net Zero 3 initiative in drive for carbon neutrality. <https://www.fm.gov.om/en/19233/>
- Ministry of Foreign Affairs, Sultanate of Oman. (2025). Oman aiming for 30% of electricity from renewables by 2030. <https://www.fm.gov.om/en/25883/>
- Ministry of Foreign Affairs, Sultanate of Oman. (2026). Oman expands renewable energy projects to strengthen energy security and sustainability. <https://www.fm.gov.om/en/38872/>
- Nama Power and Water Procurement Company. (2025). Power 7 Year Statement 2025–2031, Issue 19. <https://omanpwp.om/storage/files/1/7-year-statement/696087c603ae3.pdf>
- Oman Electricity Transmission Company. (2025). Green Finance Framework. <https://www.omangrid.com/>
- Oman Sustainability Week. (2025). OETC issues \$750 million worth green sukuk. <https://omansustainabilityweek.com/newfront/news/oetc-issues-s750-million-worth-green-sukuk>
- Reuters. (2024, January 11). Oman unveils sustainable finance framework in green push. <https://www.reuters.com/sustainability/sustainable-finance-reporting/oman-unveils-sustainable-finance-framework-green-push-2024-01-11/>
- Ulfah, I. F., Sukmana, R., Laila, N., & Sulaeman. (2023). A structured literature review on green sukuk (Islamic bonds): Implications for government policy and future studies. *Journal of Islamic Accounting and Business Research*, 15(7), 1118–1133. <https://doi.org/10.1108/JIABR-10-2022-0255>
- World Bank. (2025). Oman Vision 2040: A blueprint for sustainable growth and global integration. <https://blogs.worldbank.org/en/arabvoices/oman-vision-2040-a-blueprint-for-sustainable-growth-and-global-integration>

Appendix A. Preliminary Evidence Matrix

Table A1. Preliminary evidence matrix mapping policy, market, disclosure, and project-finance claims to source type, reliability status, and manuscript section.

Claim used in the paper	Primary evidence source	Evidence type	Date accessed	Used in section	Reliability note
Oman is targeting major renewable-energy expansion under Vision 2040	Official Omani communications and Vision 2040 material	Official policy / government source	17 May 2026	Introduction; Literature Review	High policy relevance; exact target wording must match the official document cited
Oman adopted a sustainable finance framework in 2024	Ministry of Finance material and Reuters reporting	Official framework + reputable news verification	17 May 2026	Introduction; Literature Review	Use official framework as primary; media report only as secondary confirmation
OETC issued a USD 750 million green sukuk in 2025	OETC/OSW/Baker Botts public material	Issuer / transaction announcement	17 May 2026	Green sukuk context; case motivation	Transaction details must be verified against issuer or legal-adviser releases
Avoided emissions require transparent baseline and boundary disclosure	GHG Protocol avoided-emissions guidance	International methodological standard	17 May 2026	ESG accounting; Methodology	High methodological authority; adapt carefully to project-level data availability
IFRS S1/S2 provide sustainability and climate-disclosure logic	IFRS Foundation standards pages	International reporting standard	17 May 2026	Literature Review; ESG scorecard	High authority; avoid claiming full legal adoption unless verified for Oman-specific reporting
Renewable project finance is sensitive to CAPEX, WACC, capacity factor, and revenue assumptions	IRENA cost reports and project-finance literature	Sector benchmark / analytical literature	17 May 2026	Financial modelling; sensitivity analysis	Use as benchmark evidence; final model requires project-specific calibration

Note. Reliability is judged from source type, institutional proximity to the claim, and whether secondary verification is needed. Access dates are retained for auditability, not as proof that the claim remains current. Abbreviations: ESG, environmental, social, and governance; GHG, greenhouse gas; IFRS, International Financial Reporting Standards.

Appendix B. Proposed Timeline

Table B1. Proposed twelve-month research timeline linking source validation, data collection, model construction, scenario analysis, expert validation, drafting, and final submission.

Month	Task	Output
M1-M2	Validate official sources, build evidence matrix, confirm supervisor feedback	Validated source matrix and refined research question
M2-M3	Collect policy, cost, tariff, sukuk, and benchmark data	Clean data workbook and assumptions log
M3-M4	Build DCF model and scenario controls	Financial model v1 with NPV, IRR, DSCR, and LCOE outputs
M4-M5	Construct ESG scorecard and avoided-emissions module	ESG model v1 and disclosure template
M5-M6	Compare sukuk and conventional debt assumptions	Comparative finance table and WACC/profit-rate assumptions
M6-M7	Run sensitivity and scenario analysis	Scenario outputs, tornado chart, and model validation checks
M7-M8	Conduct expert validation or document-analysis fallback	Validation memo and revised assumptions
M8-M11	Draft, revise, format, and reference-check the research paper	Full draft, supervisor revision, and final formatted manuscript
M12	Final quality control and submission	Submitted research paper or thesis chapter

Note. The timeline is indicative and should be adjusted to the university calendar, supervisor feedback, data access, and expert availability. Abbreviations: DCF, discounted cash flow; ESG, environmental, social, and governance.

Figure B1 visualises the project schedule as a Gantt chart, showing overlap, sequencing and expected deliverables across a 12-month research cycle.

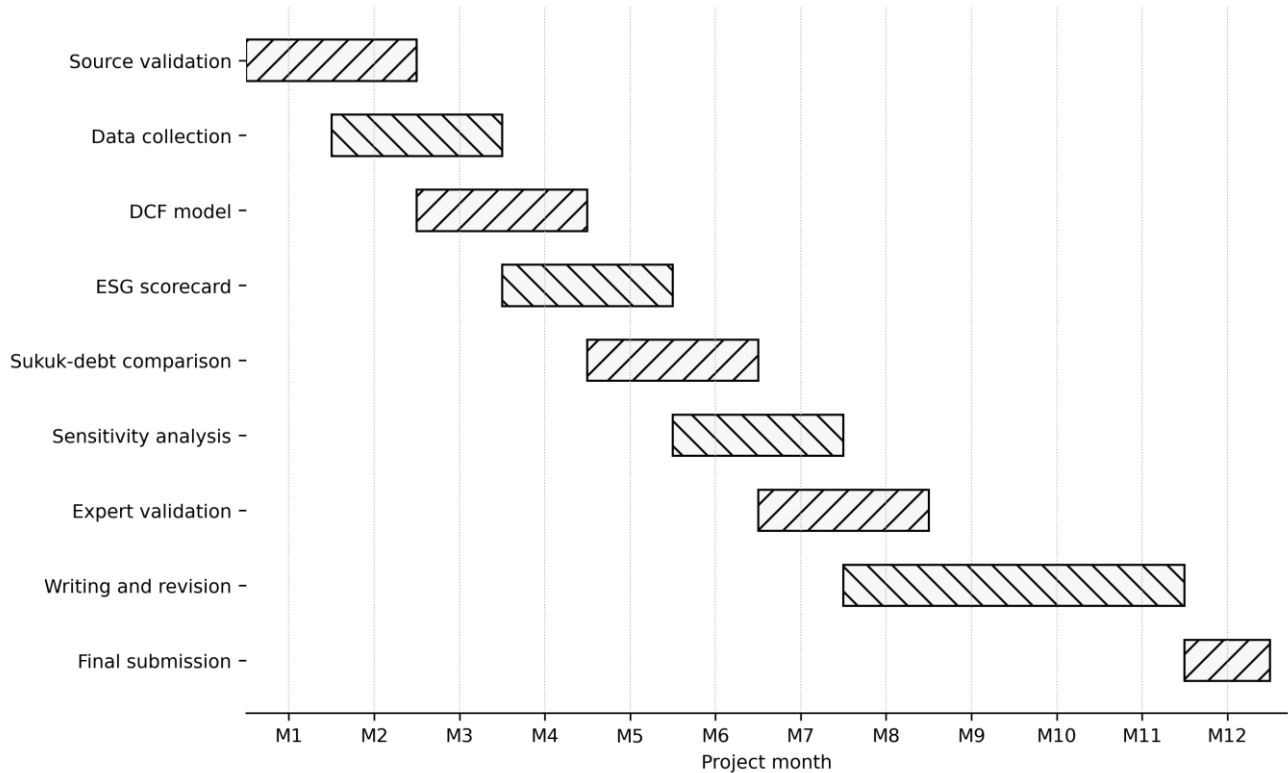


Figure B1. Twelve-month Gantt chart sequencing source validation, data collection, DCF modelling, ESG scorecard construction, expert validation, drafting, and final submission.

Note. Bars indicate the months during which each research phase is expected to be active. The schedule remains provisional until data access, expert availability, and supervisor feedback are confirmed. Abbreviations: DCF, discounted cash flow; ESG, environmental, social, and governance.

Appendix C. Semi-Structured Interview Guide

Table C1. Semi-structured interview themes, target expert groups, expected evidence, and linkage to financial-model and ESG-scorecard assumptions.

Theme	Questions covered	Expected evidence	Link to model
Green sukuk pricing and structure	Islamic finance / bank experts	Profit-rate logic, documentation burden, investor appetite, preferred Shariah structures	WACC/profit-rate assumptions and sukuk/debt comparison
Renewable-energy bankability	Energy-sector experts or utility experts	CAPEX risk, tariff certainty, curtailment, grid connection, technology bankability	DCF inputs, scenario ranges, and risk matrix
ESG credibility and avoided emissions	ESG/reporting experts	Indicator credibility, verification logic, baseline disclosure, greenwashing risks	ESG scorecard and impact-reporting framework
Policy and regulatory feasibility	Regulators/policy experts	Taxonomy, reporting template, Vision 2040 operationalisation, market-development constraints	Policy recommendations and implementation feasibility

Note. Interview evidence is used to test assumptions and identify implementation risks, not to prove project viability. Abbreviations: DCF, discounted cash flow; ESG, environmental, social, and governance; WACC, weighted average cost of capital.

For Islamic finance / bank experts

1. How do green sukuk differ from conventional green bonds in pricing, investor demand, and documentation burden?
2. Which Shariah structure would best fit renewable-energy infrastructure in Oman: Ijara, Wakalah, Musharakah, or a hybrid structure?
3. Do green labels reduce the cost of capital in practice, or do they mainly widen the investor base?

For energy-sector or utility experts

4. Which renewable technologies are most bankable in Oman under current grid and demand conditions?
5. Which financial risks dominate: CAPEX uncertainty, tariff uncertainty, curtailment, storage, or grid interconnection?
6. Which project-level data can realistically be disclosed for academic modelling?

For ESG/reporting experts

7. Which ESG indicators would carry credibility for renewable-energy projects in Oman?
8. How should avoided emissions be calculated and verified?
9. Which reporting weaknesses could create greenwashing risk for green sukuk?

For regulators/policy experts

10. Which policy measures would strengthen green sukuk issuance in Oman?
11. Should Oman adopt a dedicated taxonomy or impact-reporting template for sukuk-financed infrastructure?
12. How can Vision 2040 alignment be operationalised in financial documentation?